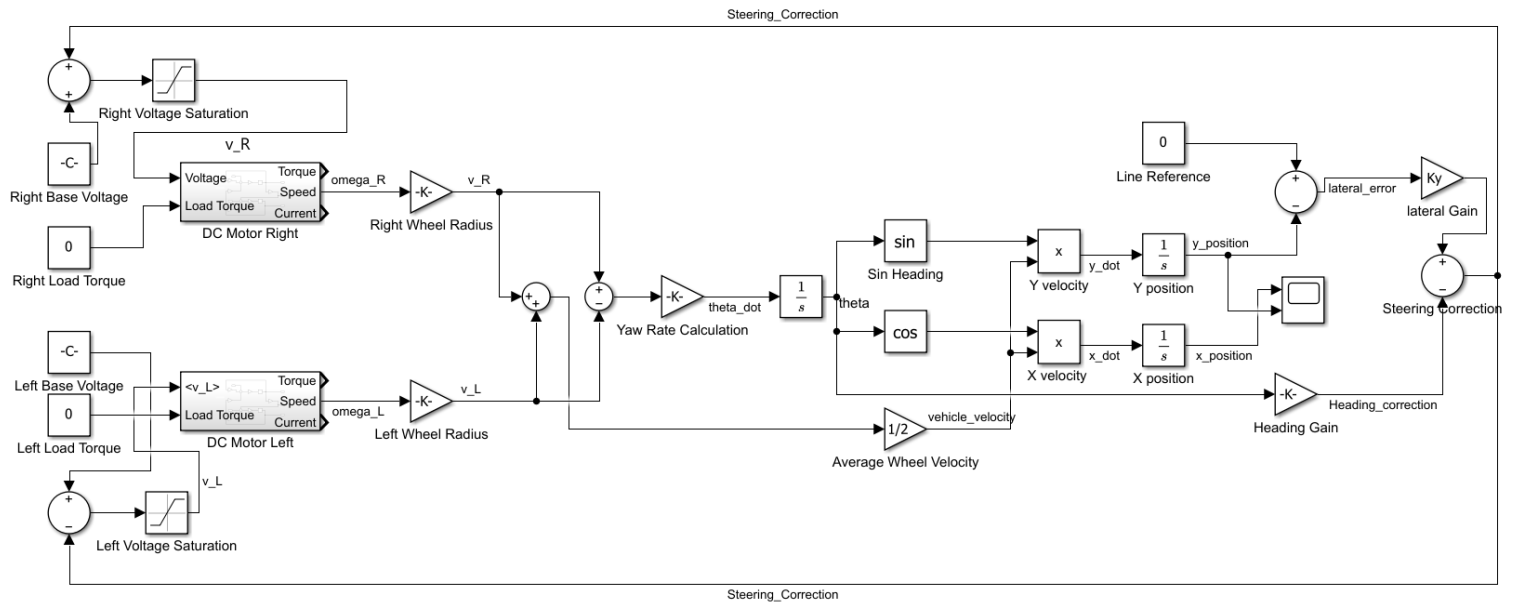


MATLAB/Simulink Modeling

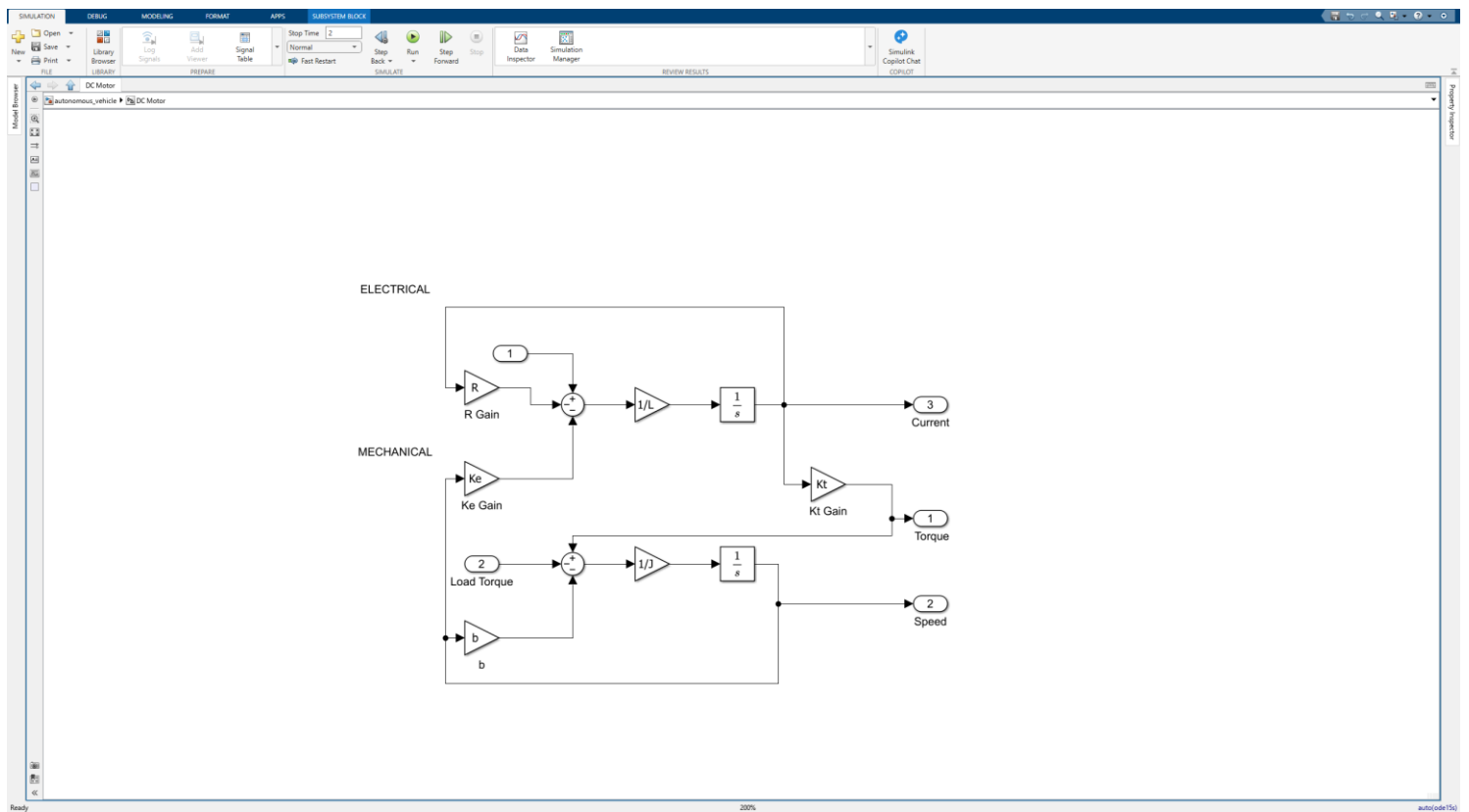
Mathematical Model of the Differential-Drive Vehicle and Line-Following Controller:

$$\begin{aligned}
 \dot{i}_R &= \frac{V_R - Ri_R - K_e\omega_R}{L} \\
 \dot{\omega}_R &= \frac{K_t i_R - b\omega_R - T_{L,R}}{J} \\
 \dot{i}_L &= \frac{V_L - Ri_L - K_e\omega_L}{L} \\
 \dot{\omega}_L &= \frac{K_t i_L - b\omega_L - T_{L,L}}{J} \\
 v_R &= r\omega_R \\
 v_L &= r\omega_L \\
 v &= \frac{v_R + v_L}{2} \\
 \dot{\theta} &= \frac{v_R - v_L}{d} \\
 \dot{x} &= v \cos \theta \\
 \dot{y} &= v \sin \theta \\
 e_y &= y_{\text{ref}} - y \\
 u &= K_y e_y - K_\theta \theta \\
 V_R &= V_{\text{base}} + u \\
 V_L &= V_{\text{base}} - u
 \end{aligned}$$

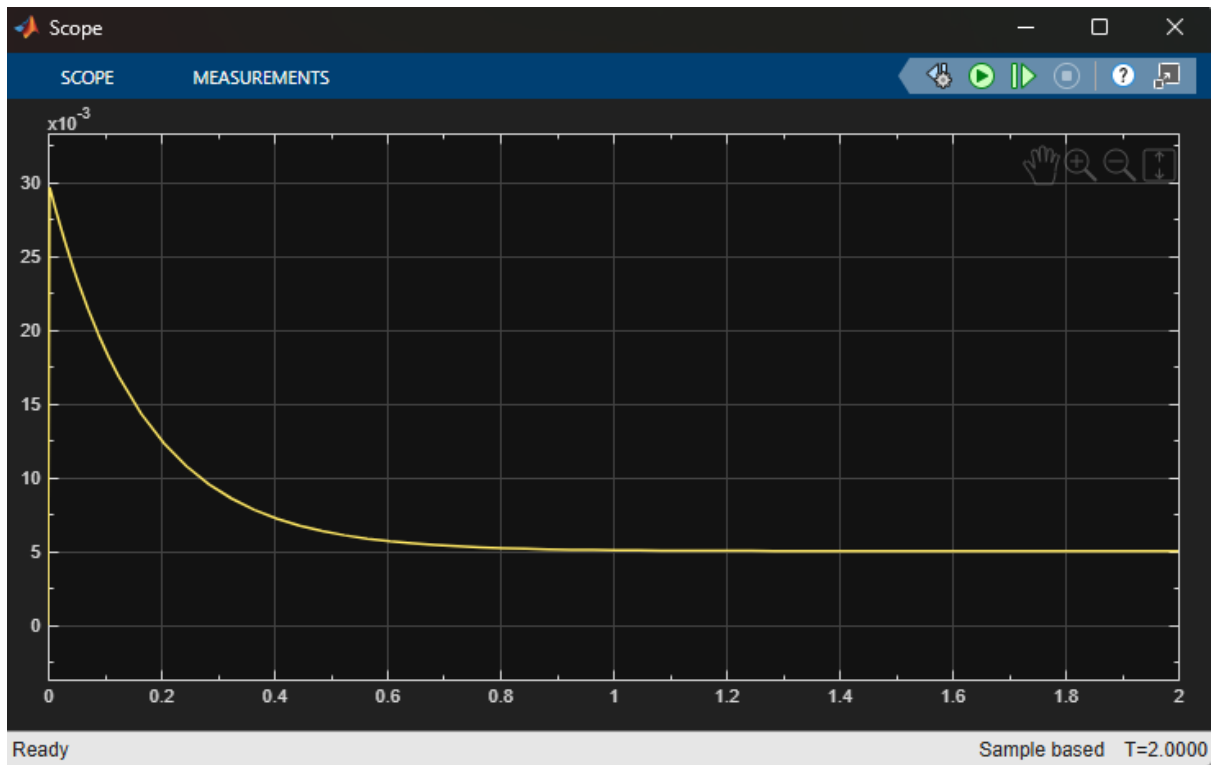
Simulink model:



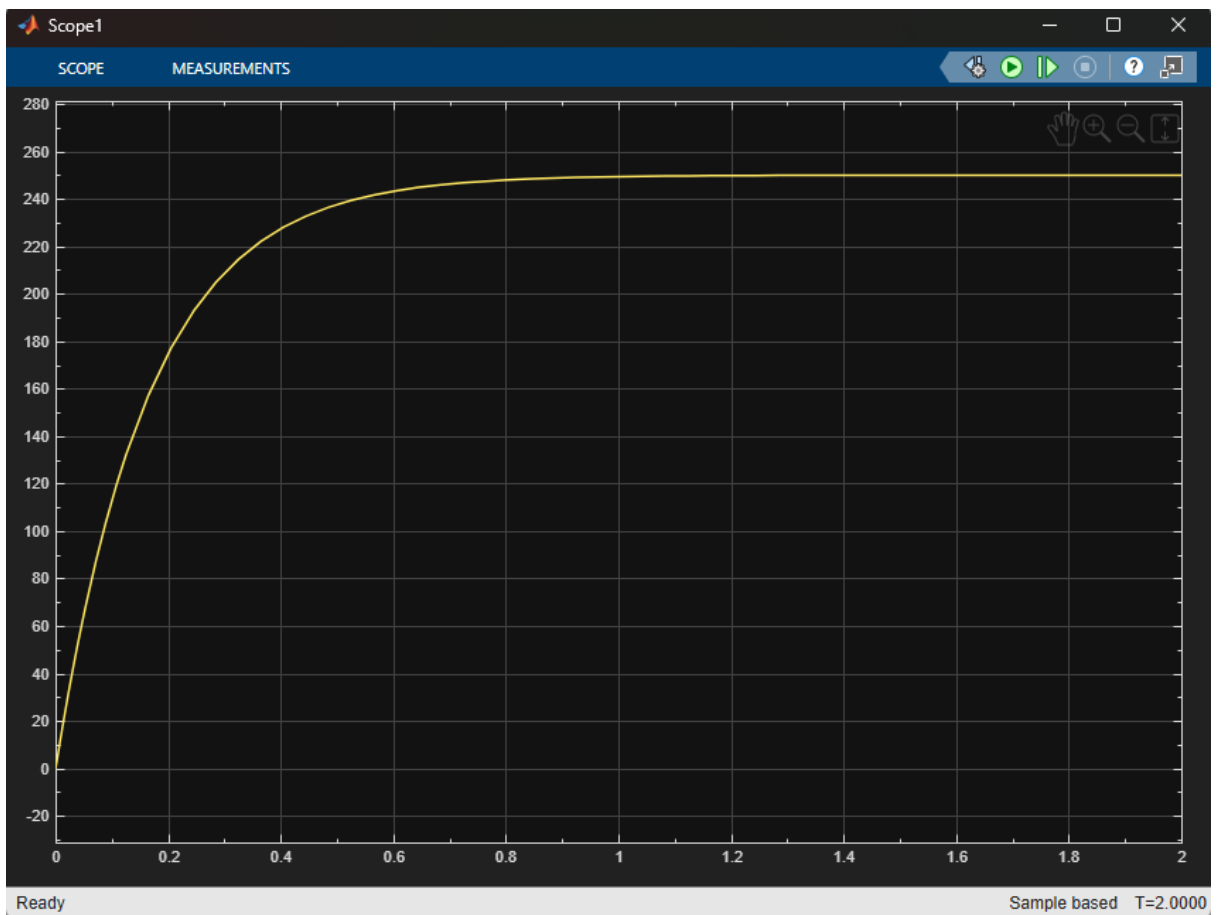
Subsystem model:



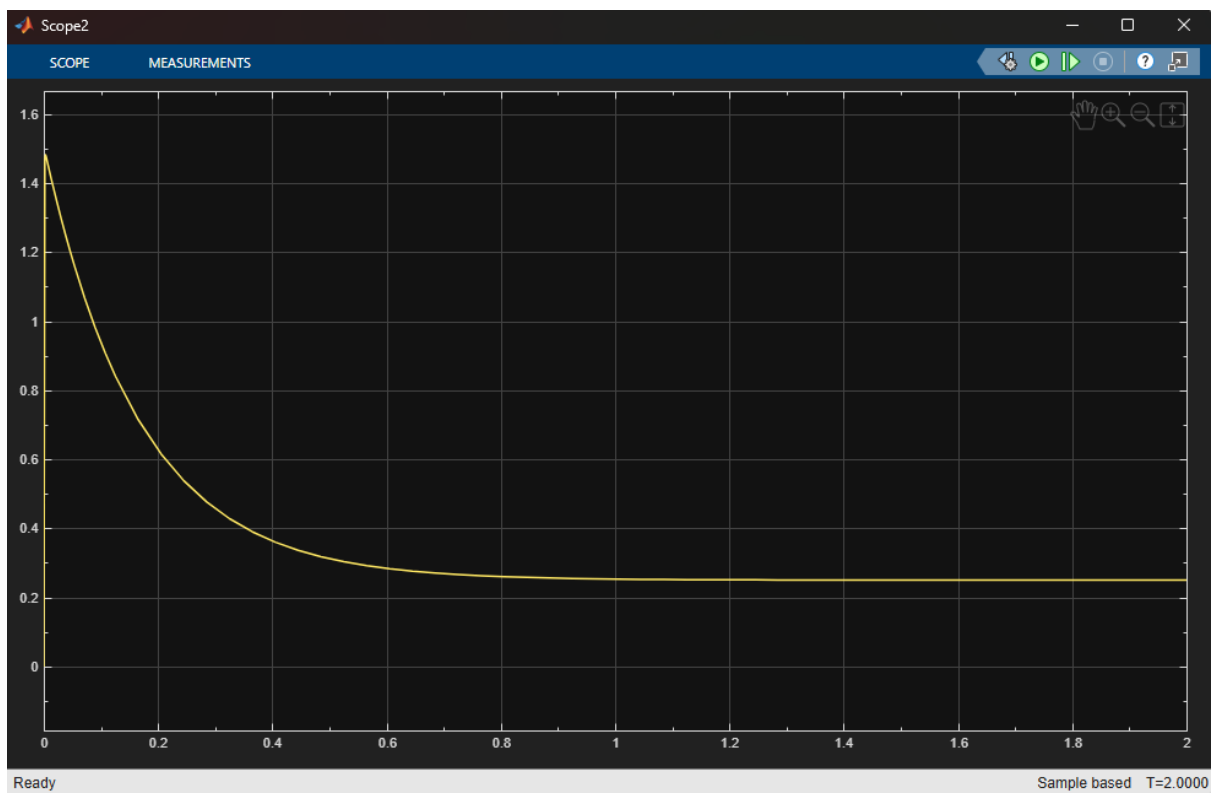
Torque Behaviour:



Speed Behaviour:



Current Behaviour:



Line following validation:

